Capstone Project Proposal

Title: Offline Library – A Walled Garden for Public Domain Works
Repository: https://github.com/stolenwise/capstone-1

Pathos

Marshall McLuhan, media theorist, famously said, "The medium is the message." With the advent of mediums like TV, smartphones, and AI, literacy is plummeting. The internet is also in digital decay. [A Pew Research Center study]() found that a quarter of all web pages that existed between 2013 and 2023 are now unreachable. This includes the accessibility to books via the internet. To protect literacy, I want to preserve the medium of books.

Project Goal

The goal of this project is to build an **offline digital library** that allows users to browse, store, and read public domain works (e.g., books in PDF or EPUB format) from a **local environment (localhost)**. The application will function as a **walled garden**, meaning that access to the content will be restricted through secure user authentication and possibly encryption. This makes the project unique from public repositories like Internet Archive by ensuring privacy and offline access, which is particularly relevant in an age of increasing content loss and digital surveillance.

Target Demographic

This tool is intended for:

- Archivists and researchers who want to store and access digital books offline.
- Readers interested in preserving and accessing public domain literature securely.
- Technically-inclined users who value **privacy, decentralization**, and **local ownership of knowledge**.

Data Source

All content in the library will be **public domain works**, initially sourced manually from trusted sites such as:

- Project Gutenberg

- Standard Ebooks
- Archive.org (where applicable and legally safe)

To avoid copyright issues, I will begin by only using works that are clearly in the public domain. Long term, I plan to build functionality to allow users to upload their own EPUBs/PDFs.

Database Schema (Rough Plan)

Initial schema in PostgreSQL:

Users

- id (PK)
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- username
- email
- password_hash
- created_at

Books

- id (PK)
- title
- author
- description
- file_path (local path to file)
- file_type (PDF/EPUB)

Sessions or AccessLogs (optional, if I add TOR/privacy features)

- id (PK)
- user_id (FK)
- timestamp

- ip_address (optional)

Functionality

Core Features

- User login/signup and authentication
- Upload public domain PDFs/EPUBs
- View library dashboard with list of uploaded books
- View or download books stored locally
- Basic file protection behind login system

Security / Privacy

- Passwords securely hashed (bcrypt or similar)
- Files accessible only after login
- Potential stretch: encrypted file storage
- TOR integration considered (not required)

User Flow

1. User signs up or logs in
2. Redirected to the library dashboard
3. User can upload new books (PDF/EPUB)
4. User can browse existing books
5. User clicks to view/download a book
6. Optional: Book tracking, notes, or favorites

Stretch Goals

- Full-text search of book contents using indexing
- File encryption & decryption with user-specific keys
- TOR integration for anonymized usage
- Tagging/categorization of books
- Read-in-browser EPUB/PDF viewer (possibly using Mozilla PDF.js)
- Dark mode for better reading experience

Anticipated Challenges

- Setting up secure file storage for sensitive documents
- PDF/EPUB compatibility and previewing in browser
- Managing file uploads and storage within Flask
- Possibly over-engineering stretch goals (like TOR or encryption)
- Keeping within scope while still exploring privacy features

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Why This Project?

As more and more digital content is lost, paywalled, or restricted, having a **local, privacy-first, user-owned** library for important public domain content becomes not just useful—but necessary. I want to explore how we can **own our digital libraries**, keep them safe offline, and access them without giving up privacy or control. This project is also an excellent way for me to deepen my understanding of full-stack web development, user authentication, secure file handling, and Flask app design.

Time Estimation

Cody (my mentor) and I evaluated the scope of the project. We believe this is realistic for the **45-65 hour** range:

- Backend (Flask, database models, routes): ~20 hours

- Frontend (basic styling and layout): ~10-15 hours
- File handling, uploads, security: ~10-15 hours
- Stretch goals: ~10-15 hours if time permits

Conclusion

This project aligns with my personal interests, values, and technical goals. It is more than CRUD—it includes secure file handling, offline privacy concerns, and unique features like local preservation of data. With a structured plan, realistic scope, and strong motivation, I believe this capstone will be challenging, rewarding, and impactful.

The password the user uses to login will allow encryption unlocking by the user's token.

So essentially it'll be double encryption. One for the file (which is done) and the one for the user when they login prevent hackers.